جامعة العلوم والتقانة
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Faculty of Engineering

Calculus and it's Applications -1

Lecture - tow

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The Limit of a Functions

Definition:

Suppose that $f(x)$ is defined when x is near the number a then we write

$$\lim_{x\to a} f(x) = L$$

and we say the limit of $f(x)$ as x approaches a equals L

Rule: $\lim_{x\to a} k = k$ *where k is const*

Properties:

1. $\lim_{x\to a} k = k$
2. $\lim_{x\to a} kf(x) = k\lim_{x\to a} f(x)$
3. $\lim_{x\to a}(f(x) + g(x)) = \lim_{x\to a} f(x) + \lim_{x\to a} g(x)$
4. $\lim_{x\to a}(f(x) . g(x)) = \lim_{x\to a} f(x) . \lim_{x\to a} g(x)$
5. $\lim_{x\to a}(\frac{f(x)}{g(x)}) = \frac{\lim_{x\to a} f(x)}{\lim_{x\to a} g(x)}$
6. $\lim_{x\to a}(f(x))^n = (\lim_{x\to a} f(x))^n$
7. $\lim_{x\to a}(\ln f(x)) = \ln(\lim_{x\to a}(f(x))$

One Sided Limits

If $f(x)$ approaches L as x approaches a through values less than a $(x < a)$
then we say that L is a limit of $f(x)$ from the left and written

$$\lim_{x\to \bar{a}} f(x) = L$$

If $f(x)$ approaches L as x approaches a through values greater than a $(x > a)$
then we say that L is a limit of $f(x)$ from the right and written

$$\lim_{x\to a^+} f(x) = L$$

$f(x) = L$ if and only if $\lim_{x\to a} f(x) = L$

Rule : we say that the limite fund if $\lim_{x\to \bar{a}} f(x) = \lim_{x\to a^+} f(x)$

Example (1):

Find:

(1) $\lim_{x\to 0} x^2 + 4x - 5$ (2) $\lim_{x\to 1} \sqrt{2x+4}$ (3) $\lim_{x\to 0} \cos x + x^2$

(4) $\lim_{x\to 0} \ln(x+7)$ (5) $\lim_{x\to 3} \frac{x^2+1}{x-2}$ (6) $\lim_{x\to 0} 9$

Solution

(1) $\lim_{x\to 0} x^2 + 4x - 5 = 0^2 + 4(0) - 5 = -5$

(2) $\lim_{x\to 1} \sqrt{2x+4} = \sqrt{2+4} = \sqrt{6}$

(3) $\lim_{x\to 0} \cos x + x^2 = \cos 0 - 0^2 = 1 - 0 = 1$

(4) $\lim_{x\to 0} \ln(x+7) = \ln(0+7) = \ln 7$

(5) $\lim_{x\to 3} \frac{x^2+1}{x-2} = \frac{9+1}{3-2} = 10$

(6) $\lim_{x\to 0} 9 = 9$

Example (2):

If $f(x) = \begin{cases} 2x-1, & x \geq 1 \\ x^2, & x < 2 \end{cases}$ Find:

(1) $\lim_{x\to 1^+} f(x)$ (2) $\lim_{x\to 1} f(x)$ (3) $\lim_{x\to 1} f(x)$

Solution

(1) $\lim_{x\to 1^+} f(x) = \lim_{x\to 1} 2x - 1 = 2 - 1 = 1$

(2) $\lim_{x\to 1^-} f(x) = \lim_{x\to 1^-} x^2 = 1^2 = 1$

(3) $\lim_{x\to 1} f(x) = 1$ $\left(\lim_{x\to 1^+} f(x) = \lim_{x\to 1^-} f(x)\right) = 1$

Exercise – 3

1. Find the domain of definition of the following functions

(1) $f(x) = x^2 - 3x + 9$ (2) $f(x) = \cot x - \tan x$

(3) $f(x) = \sqrt{4 - 2x^2}$ (4) $f(x) = \ln(6 - 5x + x^2)$

(5) $f(x) = \sqrt{|2x - 5|}$ (6) $f(x) = \frac{2x}{x^2 - 2x + 1}$

2. Which of the following functions are even and which are odd

Indeterminate Forms

Indeterminate forms $\frac{0}{0}$, $\frac{\infty}{\infty}$, $0 \times \infty$, $\infty - \infty$, 1^{∞}

theorems or mathematical operations, for example factoring, multiplying by the

conjugate, dividing numerator and denominator by the highest power of x.

Note: $\frac{A}{0} = \infty$, $\frac{0}{A} = 0$, $\frac{A}{\infty} = 0$, $\frac{\infty}{A} = \infty$, $A \equiv$ any number

Example:

Rule:

$x^2 - a^2 = (x-a)(x+a)$, $(x \pm a)^2 = x^2 \pm 2ax + a^2$,

$(x \pm a)^2 \neq x^2 \pm a^2$, $x^3 \pm a^3 = (x \pm a)(x^2 \mp ax + a^2)$,

$x^2 + a^2 \neq (x-a)(x+a)$

Find

(1) $\lim_{x\to 2} \frac{x^2-4}{x-2}$ (2) $\lim_{x\to 2} \frac{x-2}{2-x}$ (3) $\lim_{x\to 2} \frac{x^2-4x+4}{x-2}$ (4) $\lim_{x\to 3} \frac{x^3+2}{x^2-9}$

(5) lim $\lim_{x\to 1} \frac{x^2-2x+1}{x^2-1}$ (6) $\lim_{x\to 2} \frac{x^2-4}{\sqrt{x+2}-2}$

Solution

(1)

$\lim_{x\to 2}\frac{x^2-4}{x-2}=\frac{0}{0}$ $\Rightarrow \lim_{x\to 2}\frac{(x-2)(x+2)}{x-2}=\lim_{x\to 2}(x+2)=4$

(2) $\lim_{x\to 2}\frac{x-2}{2-x}=\frac{0}{0}$ $\Rightarrow \lim_{x\to 2}\frac{-(2-x)}{2-x}=\lim_{x\to 2}-1=-1$

(3)$\lim_{x\to 2}\frac{x^2-4x+4}{x-2}=\frac{0}{0}=$ $\Rightarrow \lim_{x\to 2}\frac{(x-2)(x+2)}{x-2}=\lim_{x\to 2}(x+2)=4$

(4) $\lim_{x\to 3}\frac{x^3+2}{x^2-9}=$ $\frac{29}{0}=\infty$

(5)

$\lim_{x\to 1}\frac{x^2-2x+1}{x^2-1}=\frac{0}{0}=\lim_{x\to 1}\frac{(x-1)(x-1)}{(x-1)(x+1)}=\lim_{x\to 1}\frac{x-1}{x+1}=\frac{0}{2}=0$

$\lim_{x\to 2}\frac{x^2-4}{\sqrt{x+2}-2}=\frac{0}{0}=\lim_{x\to 2}\frac{x^2-4}{\sqrt{x+2}-2}\frac{\sqrt{x+2}+2}{\sqrt{x+2}+2}=\lim_{x\to 2}\frac{(x-2)(x+2)\sqrt{x+2}+2}{x+2-4}=$

$=\lim_{x\to 2}(x+2)\sqrt{x+2}=16$

Exercise

Find:

(1) $\lim_{x\to\infty}\frac{x^3-x+5}{x^2-2x+6}$ (2) $\lim_{x\to 7}\frac{\sqrt{x+2}-3}{x-7}$ (3) $\lim_{x\to\infty}\frac{4x-3}{x^5-7x+1}$

(4) $\lim_{x\to 1}\frac{(x+1)^2-4}{x-1}$ (5) $\lim_{x\to 0}\frac{x}{\sqrt{x}}$ (6 $\lim_{x\to\infty}\sqrt{x+1}-x$

Special limits:

Cauchy theorem: $\lim_{x\to a}\frac{x^n-a^n}{x-a} = a^{n-1}$ and

$\lim_{x\to a}\frac{x^n-a^n}{x^m-a^m} = \frac{n}{m}a^{n-m}$

The limit of:

$$\lim_{x\to 0}\frac{sinx}{x} = 1 \quad and \quad \lim_{x\to 0}\frac{x}{sinx} = 1 \; and \quad \lim_{x\to 0}\frac{tanx}{x} = 1$$

The number $\boldsymbol{e}$:

$$\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^x = e, \; \lim_{x\to 0}(1+x)^{\frac{1}{x}} = e,$$

$$\lim_{x\to\infty}\left(1+\frac{a}{x}\right)^x = e^a, \; \lim_{x\to 0}(1+ax)^{\frac{1}{x}} = e^a,$$

Example: finf:

(1) $\lim_{x\to 3}\frac{x^3-27}{x-3} = \lim_{x\to 3}\frac{x^3-3^3}{x-3} = 3^{3-1} = 8$

(2) $\lim_{x\to 3}\frac{x^3-27}{x^2-9} = \lim_{x\to 3}\frac{x^3-3^3}{x^2-2^2} = \frac{3}{2}3^{3-2} = \frac{9}{2}$

(3) $\lim_{x\to\infty}\left(1+\frac{1}{x}\right)^{3x} = \lim_{x\to 0}\left(\left(1+\frac{1}{x}\right)^x\right)^3 = e^3$

(3) $\lim_{x\to 0}(1-4x)^{\frac{1}{x}} = e^{-4},$

(4) $\lim_{x\to 0}(1+x)^{\frac{5}{x}} = \left(\lim_{x\to 0}(1+x)^{\frac{1}{x}}\right)^5 = e^5$

(5)$\lim_{x\to 0}\frac{x^2+sinx}{x} = \lim_{x\to 0}\left(\frac{x^2}{x}+\frac{sinx}{x}\right) = \lim_{x\to 0}x + \lim_{x\to 0}\frac{sinx}{x} = 0+1 =$

(6)

$\lim_{x\to 0}\frac{sin3x}{x} = \frac{0}{0} \Rightarrow \lim_{3x\to 0}\frac{sinx}{x}\times\frac{3}{3} = 3\lim_{3x\to 0}\frac{sin3x}{3x} = 3\times 1 = 3$

Exercise

Find the limites:

(1) $\lim_{x\to 0} x\ cotx$ (2) $\lim_{x\to 0} x\ secx$ (3) $\lim_{x\to 0} \frac{sin^2 x}{x^2}$

(4) $\lim_{x\to 0} \frac{sinx}{5x}$ (5) $\lim_{x\to \infty} \left(1+\frac{1}{x}\right)^{3x-4}$

(6) $\lim_{x\to \infty} \left(\frac{2-2x}{1+x}\right)^{3x-4}$